Hawler Medical University

College of Medicine · Basic Science Department

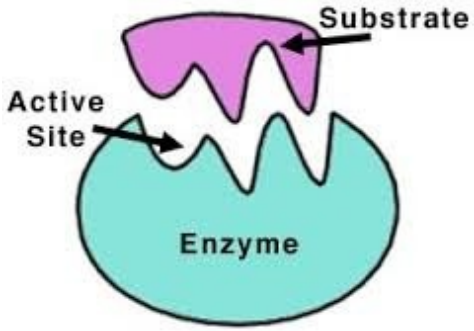
2nd Year Medical Students · 2026

Enzymes: Mechanisms, Kinetics & Clinical Applications

Clinical Biochemistry

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Learning Outcomes

- Define and classify enzymes, identifying their biological and diagnostic importance.
- Describe enzyme specificity, active sites, catalytic mechanisms, and cofactors.
- Explain how temperature, pH, and enzyme and substrate concentrations affect activity.
- Interpret Michaelis–Menten kinetics, including Km and Vmax.
- Distinguish major mechanisms of enzyme inhibition and regulation.
- Discuss the clinical significance of AST, ALT, ALP, and ACP.

What Are Enzymes ?

Enzymes are biological catalysts, mostly proteins, that speed up chemical reactions in living organisms without being consumed or permanently changed during the reaction.

Here are the properties

Not consumed: Not used up or destroyed in the reaction.
Not permanently changed: Returns to its original shape after helping.
Ready for next reaction: Instantly available to work again.
Reusable: One enzyme can repeat the same job thousands of times.

Why enzymes matter in the clinical laboratory

Biochemical Role

- Catalysts accelerate reactions without being consumed.
- Most enzymes are proteins with substrate-specific active sites.
- Activities change with pH, temperature and substrate availability.

Diagnostic Role

- Cell injury can release intracellular enzymes into blood.
- Serum activity reflects release, clearance and tissue distribution.
- A raised enzyme is a clue—not a stand-alone diagnosis.

Enzyme Classification

Class	Enzyme type	Function	Example
1	**Oxidoreductases**	Catalyze oxidation–reduction reactions	Lactate dehydrogenase (LDH)
2	**Transferases**	Transfer functional groups between molecules	ALT, AST
3	**Hydrolases**	Break chemical bonds using water	Amylase, lipase, ALP
4	**Lyases**	Remove or add groups without hydrolysis or oxidation	Decarboxylase
5	**Isomerases**	Rearrange atoms within a molecule	Phosphoglucose isomerase
6	**Ligases**	Join two molecules, usually using ATP	DNA ligase
7	**Translocases**	Catalyze movement of ions or molecules across membranes	Na^+/K^+-ATPase

Coenzymes and cofactors

Organic Coenzymes

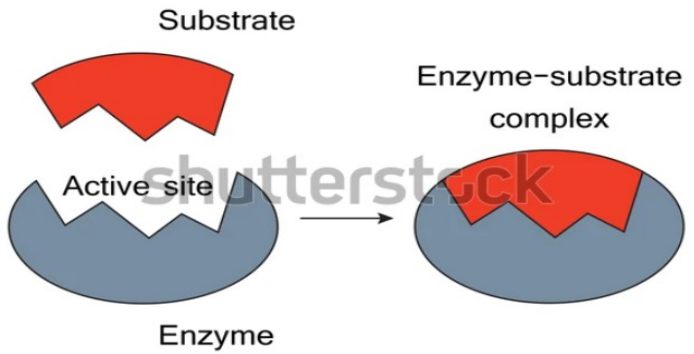
- PLP (vitamin B6): required for ALT and AST transamination.
- NAD^+/NADH (vitamin B3): used in coupled dehydrogenase assays.
- $FAD/FADH_2$ (vitamin B2) and TPP (vitamin B1): support other enzyme reactions.

Inorganic Cofactors

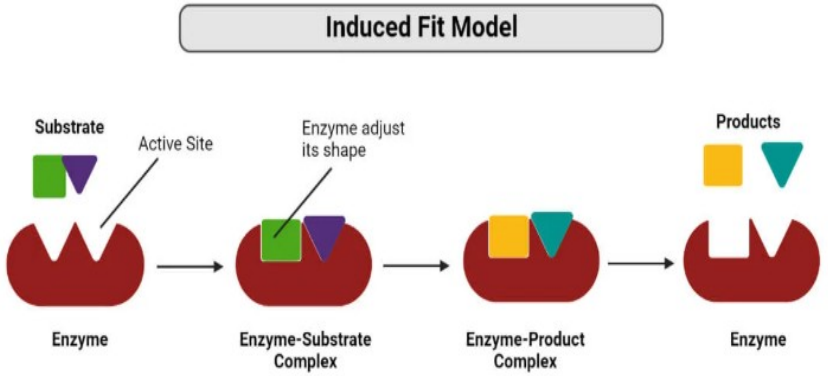
- Mg^{2+}: important in ATP-dependent enzyme systems.
- Zn^{2+}: catalytic cofactor in several enzymes.
- Holoenzyme = apoenzyme + required cofactor.

Binding Models and Catalysis

- Lock-and-Key Model: The active site has a fixed shape complementary to the substrate.



- Induced-Fit Model: Substrate binding triggers a conformational change in the enzyme, improving binding and catalysis.



Active Sites and Enzyme Catalysis

• **Acid–Base Catalysis:** Amino acid residues donate or accept protons to stabilize reaction intermediates. Histidine is effective near physiological pH.

• **Covalent Catalysis:** A nucleophilic residue forms a temporary covalent bond with the substrate

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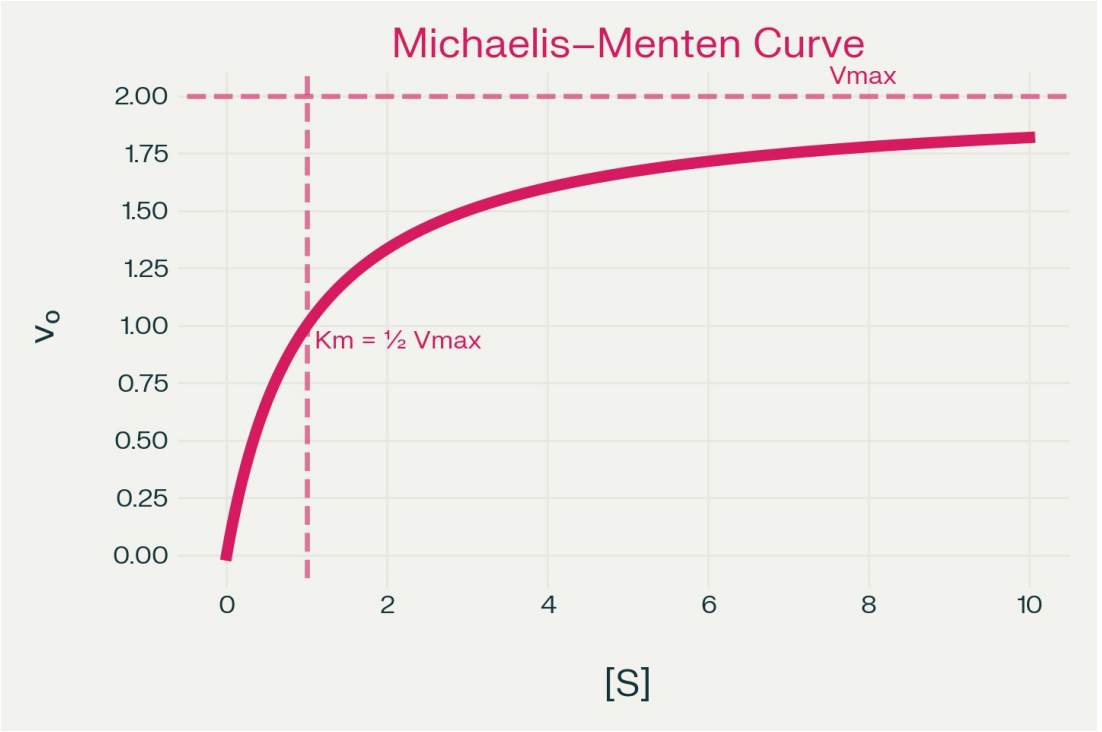
Michaelis–Menten Kinetics

Enzyme activity depends on substrate concentration and is described by the Michaelis–Menten equation.

$v = V_{max} \times [S] / (K_m + [S])$

Km is the substrate concentration at ½ Vmax.

Vmax is the limiting velocity at enzyme saturation.



Inhibition and Enzyme Regulation

Kinetic Inhibition

- Competitive: Km increases; Vmax unchanged (simple reversible model).
- Pure noncompetitive: Vmax decreases; Km unchanged.
- Irreversible inhibitors permanently reduce active enzyme.

Cellular Regulation

- Allosteric effectors modify conformation and activity.
- Zymogens are activated by proteolytic cleavage.
- Phosphorylation / dephosphorylation alters enzyme activity.

Active site and Catalytic mechanisms

How Catalysis Works

- Enzyme + substrate ⇌ enzyme–substrate complex → product.
- Induced fit: binding can alter enzyme conformation.
- Acid–base and covalent catalysis lower activation energy.

Why This Matters

- Inhibitors can reduce reaction rates and alter assay results.
- Temperature and pH are standardized in diagnostic assays.
- Enzymes are often measured by rate, not protein mass.

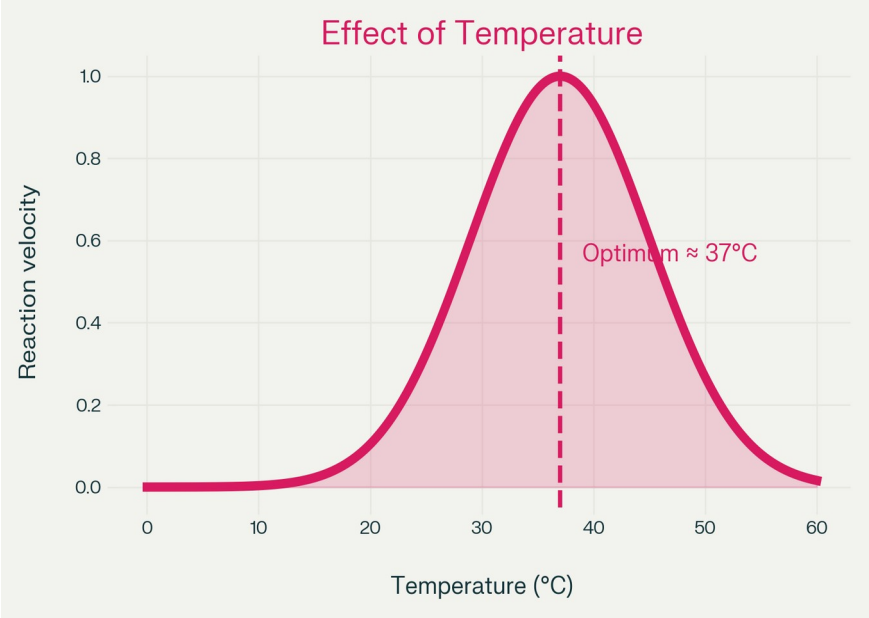
Factors Affecting Enzyme Activity

Enzyme activity is affected by several factors, including:

- Temperature
- pH
- Enzyme concentration
- Substrate concentration

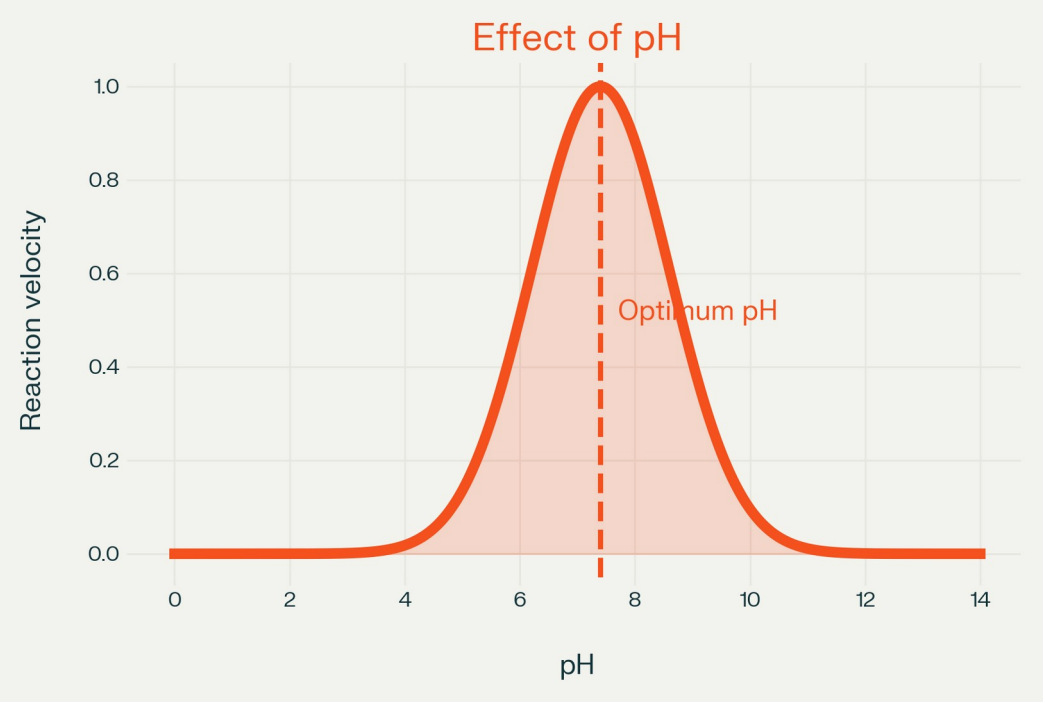
Effect of Temperature

Enzyme activity increases with temperature up to an optimum (≈37°C in humans), then declines due to denaturation. Fever and hypothermia can affect enzyme activity.



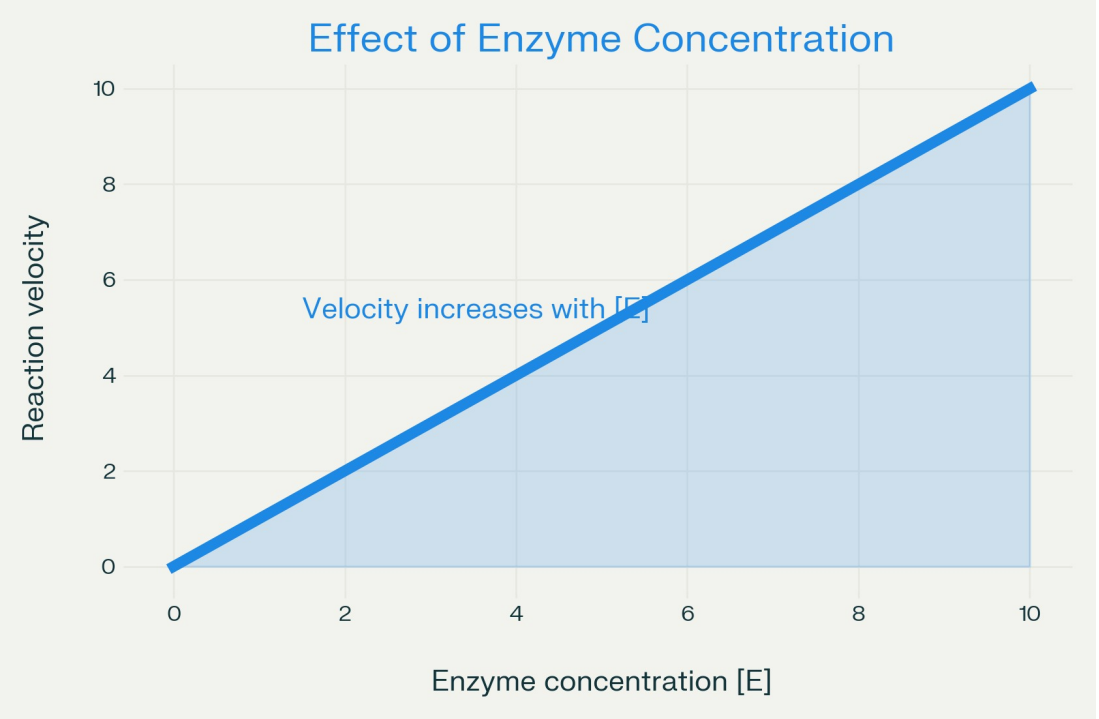
Effect of pH

Enzyme activity depends on pH, with each enzyme having an optimum. Deviations reduce activity. Pepsin works best in acidic pH, while trypsin prefers alkaline pH.



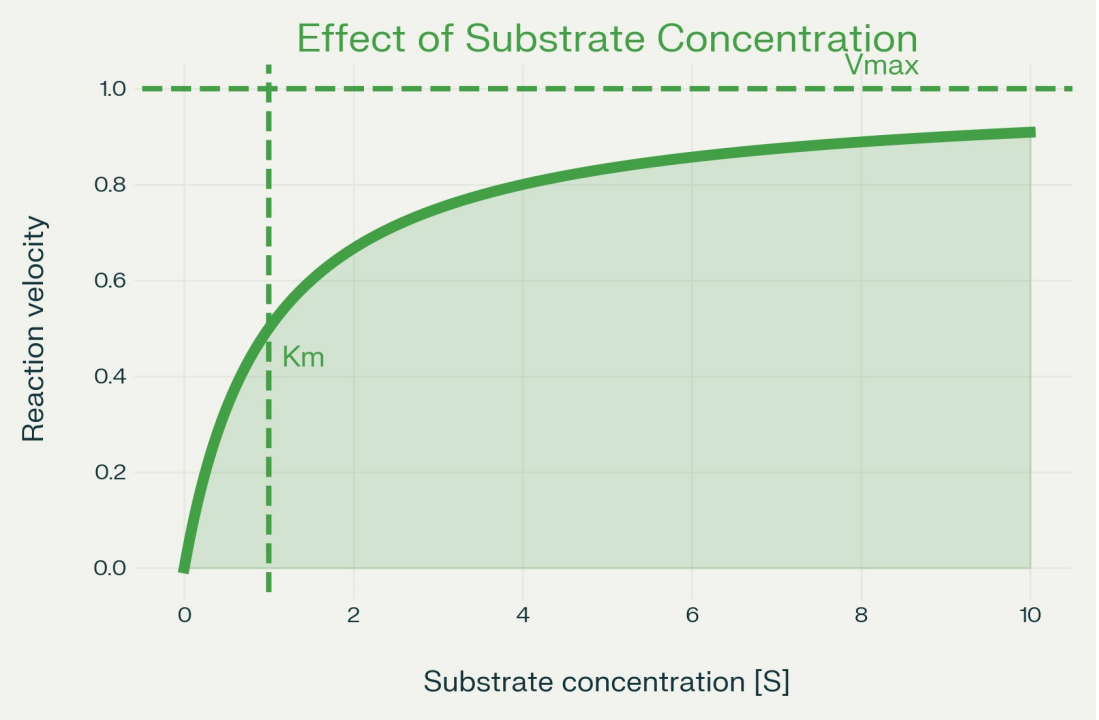
Effect of Enzyme Concentration

When substrate is sufficient, increasing enzyme concentration produces a linear increase in reaction velocity by providing more active sites. This principle is important in clinical enzyme assays.



Effect of Substrate Concentration

Reaction velocity increases with substrate concentration until enzyme saturation, approaching Vmax. Km is the substrate concentration at half Vmax.



1. Transaminases
2. Phosphatases

1. Transaminases :

The transaminases are enzymes involved in transfer of an amino group from an α-amino acid to an α-keto acid as a result, different amino acid and different keto acid are produced, which needed Cofactor (pyridoxal phosphate). All naturally occurring α-amino acids can take part of such reactions.

Clinically, there are two types of transaminases:

a) Aspartate transaminase (AST) or
Glutamate oxaloacetate transaminase (GOT)

b) Alanine transaminase (ALT) or
Glutamate pyruvate transaminase (GPT)

Aspartic acid + α-keto glutaric acid $\xrightarrow{\text{AST}}$ Oxalic acid + Glutamic acid

Alanine + α-keto glutaric acid $\xrightarrow{\text{ALT}}$ Pyruvic acid + Glutamic acid

Diagnostic application:

GOT: The major diagnostic applications are in the investigation of myocardial infarction, liver disease, and muscle disease.

GPT: Is involved in the investigation of liver disease.

Normal Range:

GOT: 4-17 I.U/L

GPT: 3-15 I.U/L

Causes of increased AST & ALT

	AST	ALT
Circulatory failure with shock and hypoxia	-√-	-√-
Liver disease	-√-	-√-
Cardiac disease	-√-	----
Physiological (neonates)	-√-	-√-
Cholestatic jaundice	-√-	-√-
Skeletal muscle disease	-√-	-√-
After trauma or surgery	-√-	-√-
severe anemia	-√-	----

2. Phosphatases

Phosphatases are enzymes that catalyze the splitting off of phosphoric acid from mono phosphoric ester.

Two types are commonly estimated in serum:

a- Alkaline phosphatase (ALP)
b- Acid Phosphatase (ACP)

Alkaline Phosphatase



Phosphate monoester

Alcohol

Phosphate

Acid Phosphatase



Phosphate monoester

Alcohol

Phosphate

Diagnostic application:

ALP: The main diagnostic applications for estimation are liver disease (cholestatic) and bone disease.

ACP: is used to detect prostatic carcinoma.

Normal range:

ALP: 21-92 I.U/L

ACP: 7-25 I.U/L

Causes of raised ALP activity:

1. Physiology

a) children /young (active bone growth)

b) pregnancy

2. Liver disease (hepatobiliary disease): like cirrhosis, hepatitis

3. Bone disease: like osteomalacia, rickets.

4. malignancy like bone tumors, liver tumors

Causes of increased serum ACP activity

1- Carcinoma of the prostate

2- Paget's disease of bone

3- Bone cancer

4- Hyperparathyroidism

References

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thank you